Reasoning test

For admission to Class 9

About the test

- 30 questions in 30 minutes: 10 each of maths, English and science, mixed together.
- Half the maths and science questions are from the Class 8 syllabus (the chapters listed below); the rest test reasoning. Each question is marked Basics or Class 8.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 2 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

What the test covers: basics

Basics questions test reasoning, not memory, on these topics.

Maths

1. Number patterns and series
2. Primes, factors, HCF and LCM
3. Fractions
4. Perimeter and area
5. Simple equations
6. Speed, distance and time
7. Averages
8. Ratio and proportion
9. Percentages, profit and loss, interest
10. Squares and cubes

English

1. Vocabulary
2. Grammar
3. Sentence completion and ordering
4. Reading comprehension

Science

1. Evaporation and condensation
2. Magnets
3. Light, shadows and mirrors
4. Acids, bases and indicators
5. Physical and chemical changes
6. Heat
7. Electric circuits
8. Respiration
9. Photosynthesis
10. Motion and speed
11. Food chains
12. Floating, sinking and density
13. Force and pressure

Reasoning

1. Clocks and calendars
2. Number codes and puzzles
3. Everyday units
4. Word problems
5. Time and work
6. Problems on ages
7. Mixtures
8. Analogies, odd one out, series and codes
9. Logical deduction
10. Reading an experiment
11. Applying a principle to a new situation

Class 8 syllabus questions

Maths

1. A Square and A Cube
2. Power Play
3. A Story of Numbers
4. Quadrilaterals
5. Number Play
6. We Distribute, Yet Things Multiply
7. Proportional Reasoning-1

Science

1. The Invisible Living World: Beyond Our Naked Eye
2. Health: The Ultimate Treasure
3. Electricity: Magnetic and Heating Effects
4. Exploring Forces
5. Pressure, Winds, Storms, and Cyclones
6. Particulate Nature of Matter

Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS · BASICS
   How many seconds are there in 2.5 hours?

   A 150 **B** 900
   C 9,000 **D** 15,000

2. ENGLISH · BASICS
   Choose the word most nearly opposite in meaning to "reluctant".

   A eager **B** slow
   C afraid **D** unwilling

3. SCIENCE · BASICS
   In the food chain grass → grasshopper → frog → snake, which living thing is the producer?

   A Frog **B** Snake
   C Grasshopper **D** Grass

4. MATHS · CLASS 8
   What is the cube of 5?

   Type-in answer: _____

5. ENGLISH · BASICS
   Choose the correct word: She has been living in Pune ___ 2019.

   A from **B** since
   C for **D** by

6. SCIENCE · CLASS 8
   Some antibiotics, such as penicillin, are made from

   A Fungi **B** Viruses
   C Mosquitoes **D** Algae

7. MATHS · BASICS
   The ratio of boys to girls in a class is 4 : 5. If there are 20 boys, how many girls are there?

   A 16 **B** 20
   C 24 **D** 25

8. ENGLISH · BASICS
   When the coach said the match was "in the bag", he meant that

   A the team had packed its kit **B** the result was still uncertain
   C victory was almost certain **D** the match had been called off

9. SCIENCE · BASICS
   A card with a word printed on it is held in front of a plane mirror. In the mirror, the word appears

   A upside down **B** with left and right swapped
   C larger than on the card **D** smaller than on the card

10. MATHS · CLASS 8
    What is the value of 10^0?

    A 0 **B** 1
    C 10 **D** It has no value

11. ENGLISH · BASICS
The soldier's courage under fire was remarkable. As used here, "remarkable" most nearly means

A ordinary
B doubtful
C hidden
D striking

12. SCIENCE · CLASS 8
Malaria is spread from person to person by

A houseflies
B mosquitoes
C cockroaches
D rats

13. MATHS · BASICS
When a number is doubled and then increased by 7, the result is 31. What is the number?

Type-in answer: _____

14. ENGLISH · BASICS
Choose the correct word: Neither the captain nor the players ___ ready.

A was
B is
C were
D has

15. SCIENCE · BASICS
Compared with the air we breathe in, the air we breathe out contains

A more carbon dioxide
B more oxygen
C less water vapour
D no carbon dioxide at all

16. MATHS · CLASS 8
What do the four angles of any quadrilateral add up to, in degrees?

Type-in answer: _____

17. ENGLISH · BASICS
To "break the ice" at a meeting means to

A make people feel at ease
B damage something by mistake
C cool a drink for guests
D end a long friendship

18. SCIENCE · CLASS 8
A force of 60 N acts on an area of 3 m^2. What is the pressure, in pascals?

Type-in answer: _____

19. MATHS · BASICS
A mother and her daughter are 50 years old together. Five years ago, the mother was 7 times as old as the daughter. How old is the daughter now?

A 8
B 10
C 12
D 15

20. ENGLISH · BASICS
Arrange the sentences in a logical order. P. Asha, their leader, checked the map. Q. The trekkers realised they were lost. R. They reached camp before dark. S. She found the right trail.

A PQSR
B QSPR
C QPRS
D QPSR

21. SCIENCE · BASICS
A sharp knife cuts more easily than a blunt one pushed with the same force, because the sharp edge

A makes much more friction with what is cut
B pushes with a larger force overall
C acts on a smaller area, raising pressure
D spreads the force over a larger area

22. MATHS · CLASS 8

What do you get when you expand $(x + 3)(x + 2)$?

A $x^2 + 5x + 6$ **B** $x^2 + 6$

C $x^2 + 6x + 5$ **D** $2x + 5$

23. ENGLISH · BASICS

What comes next in the series? Z, X, V, T, ___

A S **B** R

C Q **D** P

24. SCIENCE · CLASS 8

Why can a gas be squeezed into a smaller space much more than a solid can?

A Its particles are heavier **B** Its particles do not move

C It has no particles at all **D** Its particles are far apart

25. MATHS · BASICS

A scooter worth ₹80,000 loses 10% of its value every year. What will it be worth after 2 years?

A ₹64,800 **B** ₹64,000

C ₹66,000 **D** ₹72,000

26. ENGLISH · BASICS

Ali, Bina, Dev, Esha and Tara sit in a row of five. Tara is in the middle, Ali at the left end and Bina next to Ali. Dev is not next to Tara. Who is at the right end?

A Dev **B** Esha

C Bina **D** Cannot be determined

27. SCIENCE · BASICS

A box weighing 600 N has one face of area 2 m^2 and another of 0.5 m^2. How many pascals more pressure does it exert resting on the small face than on the large face?

Type-in answer: _____

28. MATHS · CLASS 8

The ratio of boys to girls in a school is 5 : 4. There are 360 students in all. How many are girls?

A 140 **B** 180

C 160 **D** 200

29. ENGLISH · BASICS

All rifles in the store are cleaned weekly. Some weapons in the store are rifles. Which statement must be true?

A All weapons are cleaned weekly **B** Some weapons are cleaned weekly

C No weapon is cleaned weekly **D** Everything cleaned is a rifle

30. SCIENCE · CLASS 8

An electromagnet picks up ten pins. Which change would let it pick up more?

A Use a plastic core **B** Reduce the current

C Add more turns of wire **D** Switch the current off

Answers

1	**C. 9,000**	One hour has 60 × 60 = 3,600 seconds, so 2.5 hours has 2.5 × 3,600 = 9,000 seconds. 150 is the number of minutes.
2	**A. eager**	Reluctant means unwilling or hesitant, so its opposite is eager. "Unwilling" is a word with the same meaning, not the opposite.
3	**D. Grass**	A producer makes its own food by photosynthesis, and in a food chain it always comes first. Grass is the producer; the others are consumers that eat.
4	**125**	The cube of 5 is 5 × 5 × 5 = 125. 25 is its square and 15 is 5 × 3. This is a type-in question.
5	**B. since**	"Since" is used with a point in time (since 2019); "for" is used with a length of time (for five years).
6	**A. Fungi**	Penicillin was first obtained from a mould, a kind of fungus; other antibiotics come from fungi and bacteria. Viruses are not used to make them.
7	**D. 25**	20 boys is 4 parts, so one part is 5 students. Girls are 5 parts, which is 25. 16 comes from using the ratio the wrong way round.
8	**C. victory was almost certain**	"In the bag" is an idiom meaning that success is as good as secured. It has nothing to do with an actual bag.
9	**B. with left and right swapped**	A plane mirror forms an image the same size and the same way up, but with left and right swapped, which is why the word reads backwards.
10	**B. 1**	Any non-zero number raised to the power 0 is 1, because $10^n \div 10^n = 10^0$ and also equals 1.
11	**D. striking**	Remarkable means worth noticing because it is unusual: striking. "Ordinary" is close to its opposite.
12	**B. mosquitoes**	The malaria parasite is carried by the female Anopheles mosquito, which passes it on when it bites. Stagnant water lets mosquitoes breed.
13	**12**	Work backwards: 31 − 7 = 24, and 24 ÷ 2 = 12. Check: 12 × 2 + 7 = 31. This is a type-in question.
14	**C. were**	With "neither … nor", the verb agrees with the nearer subject. "Players" is plural, so the verb is "were".
15	**A. more carbon dioxide**	The body uses oxygen to release energy from food and gives off carbon dioxide, so breathed-out air holds more carbon dioxide and less oxygen.
16	**360**	A diagonal splits a quadrilateral into two triangles, each with angles adding to 180°, so the total is 360°. This is a type-in question.
17	**A. make people feel at ease**	Breaking the ice is getting people talking and relaxed when they first meet. Nothing frozen is involved.
18	**20**	Pressure is force divided by area: 60 N ÷ 3 m^2 = 20 Pa. This is a type-in question: the answer is typed as a number.

19	**B. 10**	Five years ago their ages added to 40, in the ratio 7 : 1, so they were 35 and 5. The daughter is now 10 and the mother 40.
20	**D. QPSR**	The problem comes first (Q), then the response (P), then the solution (S), then the result (R). "She" in S needs Asha named before it.
21	**C. acts on a smaller area, raising pressure**	Pressure is force divided by area. The same force on the thin edge of a sharp knife gives a much greater pressure, which cuts.
22	**A. $x^2 + 5x + 6$**	Multiply each term by each term: $x^2 + 2x + 3x + 6 = x^2 + 5x + 6$. $x^2 + 6$ forgets the two middle terms.
23	**B. R**	Each letter is two places before the one before it: Z, X, V, T, R. S moves back only one place.
24	**D. Its particles are far apart**	Gas particles have large spaces between them, so pushing them closer reduces the volume. In a solid the particles are already packed tightly.
25	**A. ₹64,800**	After one year it is worth ₹72,000. The second year takes 10% of ₹72,000, which is ₹7,200, leaving ₹64,800. ₹64,000 takes ₹8,000 off twice.
26	**A. Dev**	Seats 1, 2 and 3 are Ali, Bina and Tara. Dev and Esha take seats 4 and 5; seat 4 is next to Tara, so Dev must be in seat 5.
27	**900**	On the small face: 600 ÷ 0.5 = 1,200 Pa. On the large face: 600 ÷ 2 = 300 Pa. The difference is 900 Pa. This is a type-in question.
28	**C. 160**	There are 5 + 4 = 9 parts, each of 40 students, so the girls are 4 × 40 = 160. 200 is the number of boys.
29	**B. Some weapons are cleaned weekly**	The weapons that are rifles are cleaned weekly, so at least some weapons are. Nothing is said about the weapons that are not rifles.
30	**C. Add more turns of wire**	More turns of wire, or more current, make an electromagnet stronger. A plastic core in place of iron, or less current, weakens it.